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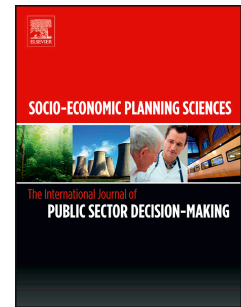
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A survey and analysis of the first 40 years of scholarly literature in DEA: 1978-2016

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Abstract

In recent years there has been an exponential growth in the number of publications related to theory and applications of Data Envelopment Analysis (DEA). Charnes, Cooper, and Rhodes (1978) introduced DEA as a tool for measuring efficiency and productivity of decision making units. DEA has immediately been recognized as a modern tool for performance measurement. Since then, a large and considerable amount of articles has been appeared, including significant breakthroughs in theory and a great portion of works on DEA applications, both public and private sectors, to assess the efficiency and productivity of their activities. Although there have been several bibliographic collections reported, a comprehensive analysis and listing of DEA-related articles covering its first four decades of history is still missing. This paper, thus, aims to report an extensive listing of DEA-related articles including theory and methodology developments and "real" applications in diversified scenarios from 1978 to end of 2016. Some summary statistics of the publications' growth, the most utilized academic journals, authorship analysis, as well as keywords analysis are also provided.

Keywords. Data Envelopment Analysis; efficiency and productivity; bibliography; survey

1. Introduction

There has been continuous and rapid growth in the field of Data Envelopment Analysis (DEA)-related studies since the original work of Charnes, Cooper, and Rhodes [Measuring

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the efficiency of decision-making units. *European Journal of Operational Research* 1978, 2(6): 429–44] (referred to CCR). This paper aims to provide full list of DEA publications since 1978. To the best of our knowledge, this listing of DEA-related articles is the most complete source of references on DEA theory and its applications in measuring the efficiency, productivity, or performance of decision-making units (DMUs). Authors hope that this new updated bibliography on DEA-related articles will be helpful to researchers and practitioners in this field when exploring the new frontiers in DEA in both theory and applications. Although we tried to include all articles, however, due to large number of publications it is not possible to guarantee a complete and accurate list with no omission of the data. Should readers happen to find any error(s) in this paper, please feel free to report them to the authors, see details at <http://DEAzone.com/biblio/>.

2. Bibliography of DEA

There are already several other bibliographies in the area of DEA which have been reported in the existing literature, *e.g.* Emrouznejad and Thanassoulis (1996a, 1996b, 1997), Seiford (1994, 1997), Tavares (2002), Gattoufi *et al.* (2004a, b), and Emrouznejad *et al.* (2008). In this study we searched related articles on SCOPUS (<http://www.scopus.com/>) and merged them into authors' dataset that is mainly gathered from the website of www.DEAzone.com. We considered publications until end of 2016. It should be noted that we only included DEA-related articles published in journals, we have ignored any articles published as chapter book or working papers. In total, there are 9881 DEA-related articles reported in this study.

As a worldwide accepted management science and operations research (MS/OR) tool, the evolution of DEA has been tracked here in terms of the increase of numbers of both articles and authors. First, it is obvious from our analysis that DEA is an emerging topic and an essential mathematical tool for measuring efficiency or performance which has received a great number attentions in diversified fields of management science. Second, three stages of the development status of DEA-related articles are detected in our study to demonstrate the tendency of DEA as an "exponential" growth. Third, several traditional MS/OR journals including *European Journal of Operational Research*, *Journal of the Operational Research*

Society, *Journal of Productivity Analysis*, *Omega* and *Annals of Operations Research* are the most utilized journals for DEA publications, while journal of *Socio-Economic Planning Sciences* has been identified first choice for DEA papers with applications in the public sector. In total, there are 94% of DEA-related articles have less than 4 authors and about 34% were published by two authors. Energy, industry, banking, educations and healthcare including hospital are found to be the most popular application areas. We detected approximately 11,961 distinct DEA authors and 25,137 distinct keywords in all DEA-related articles in our database, which has also been grown significantly in recent years.

The next section lists a series of selected descriptive statistics involving the numbers and distributions of papers, journals, authors, keywords, as well as pages within our database of DEA-related articles.

3. Statistics on DEA publications

Statistics based on different years

Figure 1 demonstrates the rapid increase of DEA-related articles by year. It should be noted that there was "exponential" growth in the number of journal articles since the seminal work of CCR in 1978.

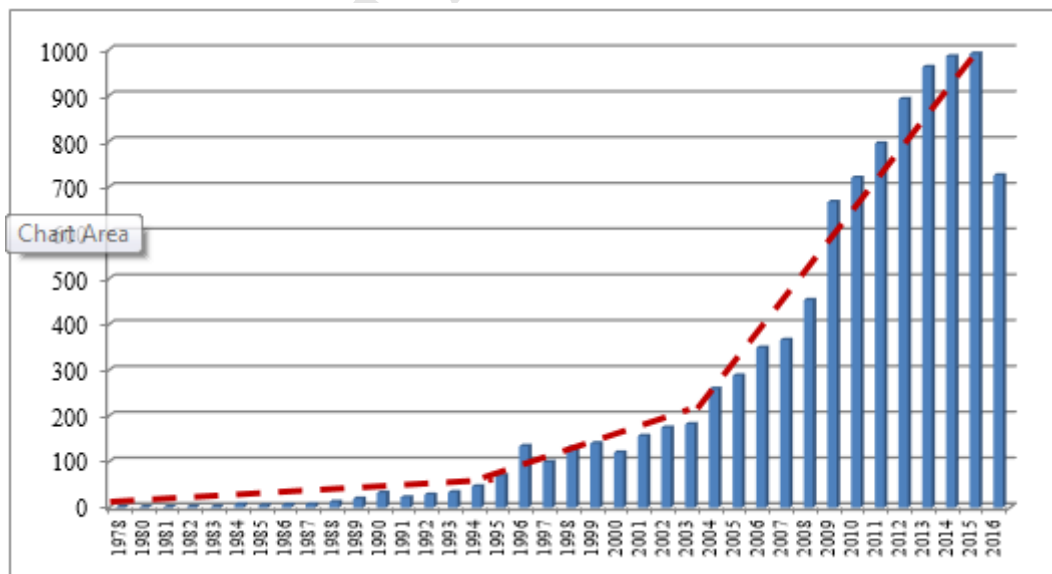


Figure 1. Distribution of DEA-related articles by year (1978-2016).

In recent four decades (1978-2016), there are totally 9881 DEA-related articles in the literature. In particular, in the last three years (2014, 2015 and 2016), the numbers of journal articles reached to just about 1000 published works in each year. In general, we can roughly classify the development status of DEA-related articles into three stages: (1) 1978-1994: in this stage the growth of DEA-related articles is relatively slow in the sense of the article numbers; (2) 1995-2003: in the second stage the growth of DEA-related articles is relatively stable and the average number of published articles is about 134 per year; (3) 2004-present: in the third stage the number of DEA-related articles show the "exponential" growth and the average number reaches about 650 per year.

Statistics based on different journals

Table 1 intends to show the top 20 journals that have published the greatest number of DEA-related articles in the last 40 years.

Table 1. Top 20 journals that have published the greatest number of DEA-related articles (1978-2016).

Row	Journal	Numbers	% of papers
1	European Journal of Operational Research	684	23.56
2	Journal of the Operational Research Society	278	9.58
3	Journal of Productivity Analysis	252	8.68
4	Omega (United Kingdom)	228	7.85
5	Expert Systems with Applications	181	6.23
6	Annals of Operations Research	167	5.75
7	Applied Economics	128	4.41
8	Socio-Economic Planning Sciences	113	3.89
9	Applied Mathematics and Computation	94	3.24
10	Energy Policy	94	3.24
11	Computers and Industrial Engineering	90	3.10
12	International Journal of Production Economics	86	2.96
13	Computers and Operations Research	74	2.55
14	Energy Economics	69	2.38
15	International Journal of Production Research	66	2.27
16	Benchmarking	62	2.14
17	Journal of Cleaner Production	62	2.14
18	Applied Mathematical Sciences	61	2.10
19	Applied Economics Letters	58	2.00
20	Applied Mathematical Modelling	56	1.93
Total		2903	100.00%

From Table 1, *European Journal of Operational Research*, *Journal of the Operational Research Society*, *Journal of Productivity Analysis*, and *Omega* are the most utilized journals. It is reasonable since DEA theory and most applications fall within the fields of management science and operations research (MS/OR), which are exactly the scopes covered by these journals.

Statistics based on different authors

In this study 11,961 distinct DEA authors were identified in total, with an average of 2.6 authors per publication. About 17% of all DEA-related articles were written by a single/sole author, while about 34% were published by two authors. Most of all DEA-related articles, about 94%, have four or less than four authors. The article with most authors has 30 distinct names. The following Figure 2 shows the distribution of DEA-related articles by number of authors.

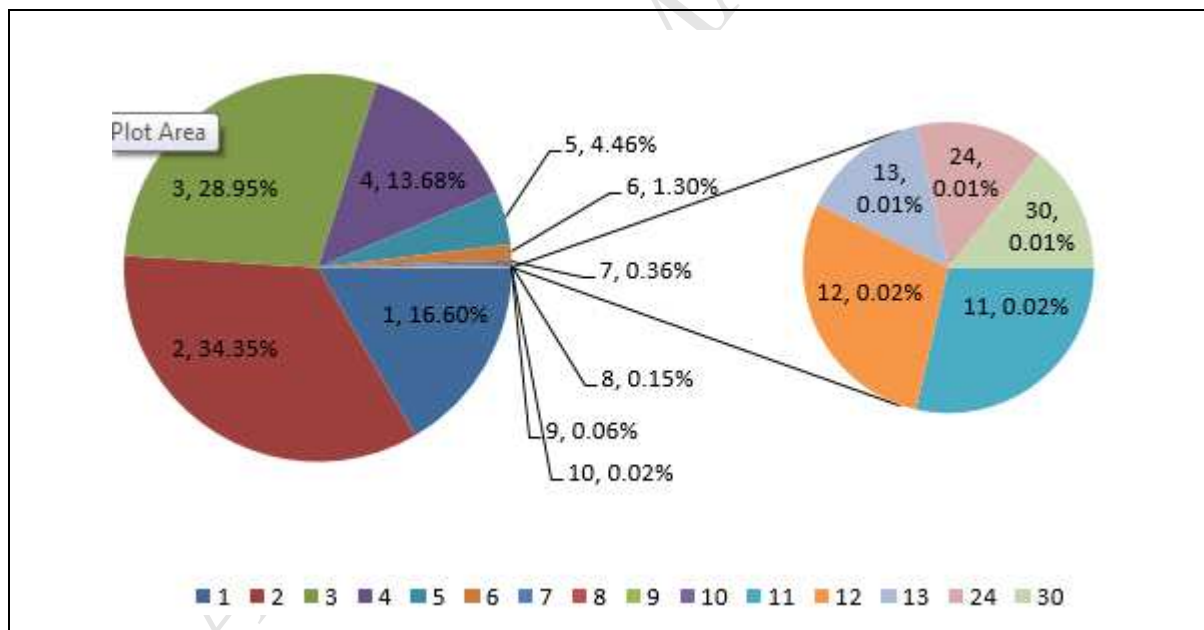


Figure 2. Distributions of DEA-related articles by number of authors (1978-2016).

Statistics based on keywords used

According to our dataset most DEA-related journal papers have identified keywords, and so approximately 25,137 distinct keywords were used. The following Table 2 shows the first 50

most used keywords. There is no doubt that Data Envelopment Analysis, DEA, or DEA models are the most used keywords with number of article of 9,577. Moreover, efficiency, decision-making, technical efficiency, linear programming, and productivity are also most popular keywords in the existing DEA-related articles.

Table 2. The 50 most used keywords in DEA-related articles (1978-2016).

No.	Keywords	No. of publications
1	Data Envelopment Analysis (DEA), Data Envelopment Analysis, DEA, or DEA models	9577
2	Efficiency	2284
3	Decision making	1005
4	Technical efficiency	840
5	Linear programming	692
6	Productivity	692
7	Mathematical models	550
8	Data reduction	513
9	Benchmarking	481
10	Decision making unit, decision-making units	459
11	Human, Humans	457
12	Efficiency measurement	422
13	Optimization	353
14	Malmquist index, Malmquist Productivity Index	344
15	Operations research	323
16	Data handling	295
17	China	286
18	Energy efficiency	274
19	Performance assessment	265
20	Regression analysis	264
21	Economics	261
22	Performance	261
23	Bootstrapping, bootstrap	261
24	Efficiency, Organizational	254
25	Returns to scale	243
26	Data analysis	234
27	Eurasia	215
28	United States	215
29	Industry	199
30	Performance evaluation	199
31	Relative efficiency	198
32	Banking	185
33	Ranking	185
34	Problem solving	176
35	Data Envelopment	175

36	Costs	170
37	Resource allocation	169
38	Efficiency analysis	166
39	Europe	162
40	Decision theory	161
41	Competition	156
42	Performance measurement	151
43	Mathematical programming	148
44	Sustainable development	145
45	Total factor productivity	143
46	Organization and management	142
47	Profitability	135
48	Numerical model	132
49	Sensitivity analysis	130
50	Methodology	125

Statistics based on page numbers (Size)

In excess of 115,277 pages of scholarly DEA-related articles have been published over the past four decades in scientific journals, with an average of 12.3 pages per article. Nearly 22% of the DEA-related articles are between 9 and 11 pages in length, with approximately 42% being between 7-12 pages. About 73% of the DEA-related articles are between 6 and 18 pages in length. The following Figure 3 shows the distribution of DEA-related articles by number of pages. After all, DEA-related articles appear to have a distribution by page numbers similar to those found in most other MS/OR fields of study.

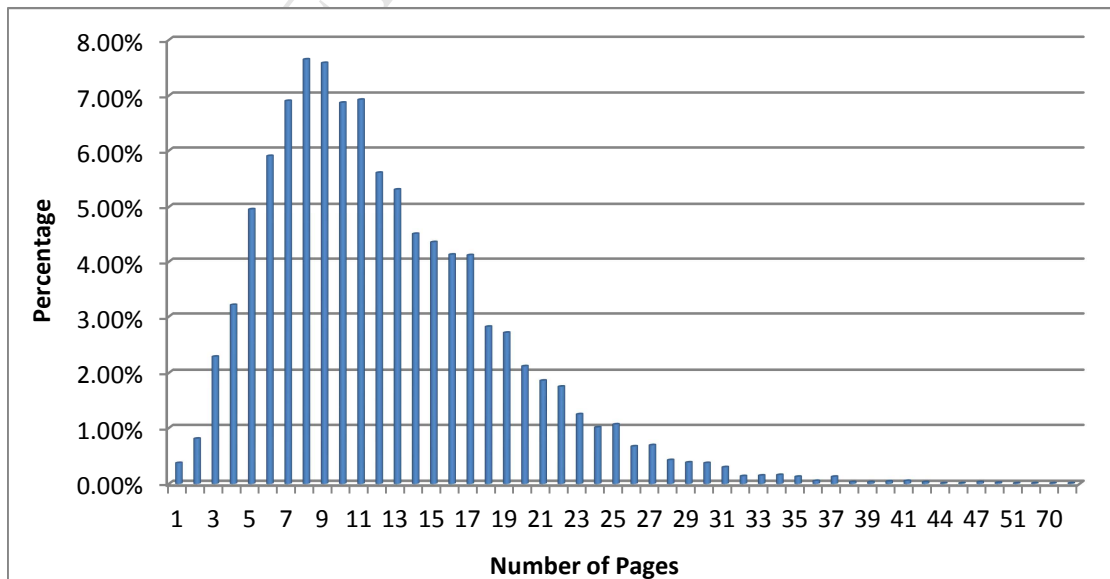


Figure 3. Distribution of DEA articles by number of pages (1978-2016)..

4. Current studies and future trends

In order to discover the current studies and future trends of DEA research, we further investigated the keywords of journal articles published in the last two years (2015 and 2016). To reflect the current studies we list, in Table 3, the top 5 most popular research keywords appeared in the DEA-related articles published in the last two years.

Table 3. Top 5 most popular research keywords in 2015 and 2016.

No	Research keywords
1	Eco-efficiency, undesirable outputs, directional distance function (DDF), environmental efficiency, carbon dioxide emissions, pollution, sustainable development, sustainability, environmental protection
2	Network DEA, two-stage DEA, efficiency Decomposition
3	Benchmarking
4	Bootstrap, bootstrapping
5	Returns to scale, scale efficiency

Next, we further list, in Table 4, the top 5 latest application fields of DEA with the greatest numbers of journal articles, including Agriculture, Banking, Supply Chain, Transportation, as well as public policy.

Table 4. Application fields of DEA methodology in 2015 and 2016.

No.	Application field
1	Agriculture
2	Banking
3	Supply chain
4	Transportation
5	Public policy

5. Conclusions and discussions

Since the seminal work of Charnes, Cooper, and Rhodes (1978) in DEA, there has been an "exponential" growth in the number of journal articles in recent four decades (1978-2016). Until end of 2016, the total number of journal articles reaches 9881 and the distinct authors reach 11,961 in total. Based on the statistics of journal articles, we found that (1) *European Journal of Operational Research*, *Journal of the Operational Research Society*, *Journal of*

Productivity Analysis, *Omega* and *Annals of Operations Research* are the most utilized journals in this field while journal of *Socio-Economic Planning Sciences* has been recognized as the first choice journal for DEA papers with applications in the public sector; (2) Most of all DEA-related articles have less than 4 authors and the percentage is 94%; (3) Data Envelopment Analysis (including DEA, or DEA models) are the most used keywords, and Efficiency, decision-making, technical efficiency, linear programming, as well as productivity are second most popular keywords in the existing DEA-related articles; (4) Environmental efficiency and DDF (including eco-efficiency, undesirable outputs, directional distance function (DDF), environmental efficiency, carbon dioxide emissions, pollution, sustainable development, sustainability, environmental protection), network DEA (including two-stage DEA, efficiency decomposition), benchmarking, bootstrap or bootstrapping, as well as returns to scale (including scale efficiency) are the main fields of current studies. Moreover, we detect agriculture, banking, supply chain, transportation, as well as public policy are the top 5 application fields of DEA with the greatest numbers of journal articles in 2015 and 2016.

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